

Module 7 Change & Verification Report

1. Traceability & Context Map

- **Target Module:** Module 7 — Knowledge-File Skill & Study Pack Architecture
- **Reconstruction Date:** 2026-09-15
- **Auditor:** PlayIQ Swarm Coordinator (`playiq-coord`) & Antigravity
- **Status:** Complete & Verified (Paired PDF compiled, Drive verified)

2. Change Summary

Section	Detail
Previous Problem	Previous draft lacked active experiments comparing context injection methods and did not connect study pack rules directly to the AI tutor project knowledge base.
Nodes Fully Restored	Node 1: The Context Window & Hallucination Defense (Garbage In, Garbage Out in LLMs). Node 2: The 8-Part PlayIQ Study Pack Architecture (Subject metadata, Core notes, 3-sentence summary, 5 key terms, 3 test questions, 1 missing link, mistake log, and 3 precision prompts). Node 3: RAG (Retrieval-Augmented Generation) mechanics explained simply for teens. Node 4: Information sanitation and privacy boundaries (zero personal data in study packs).
New Interactive Labs	Learn How You Learn Lab: Active comparison testing Messy Raw Context Dump vs. Structured 8-Part Study Pack on AI hallucination rates and accuracy.
Tutor Rule Staged	Rule 7 (`study-pack-format`): Staged in Learning Supercharger Blueprint, directing the AI tutor to enforce clean context boundaries before answering homework queries.
Knowledge File Updated	Structured YAML record appended to `student_profile.yaml` under `Tutor Knowledge Pack`.
Proof Artifact	`Complete 8-Part Study Pack` and `File Quality Test Card` for an upcoming unit in a high-priority school subject.

Mastery
Check
Challenge

The Context Master Boss Battle: Feeding both a messy notes dump and a structured study pack into an AI to detect 3 hallucinated "facts" in the messy run and verify 100% fidelity in the structured run.

3. Verification Scorecard (9-Part Non-Negotiable Quality Gates)

Component	Status	Evidence in Text
A. Lightning Hook	PASS	Part 0: The 30-second context hook (Comparing lost notes vs. indexed binder)
B. Learn How You Learn Lab	PASS	Part 3: Context Injection Trial (Messy Dump vs. Structured 8-Part Pack)
C. Active Method Comparison	PASS	Part 4: Comparative matrix tracking hallucination rate, answer precision, and study speed
D. Evidence Capture	PASS	Quantitative scores across both knowledge-file formats
E. Pattern Interpretation	PASS	Diagnostic breakdown on how structured prompts prevent AI hallucination
F. Blueprint Rule Staging	PASS	Part 6: Rule 7 (`study-pack-format`) with strict schema verification
G. Knowledge File Record	PASS	Part 7: Structured YAML calibration block for `student_profile.yaml`
H. Proof Artifacts	PASS	Part 8: Verified 8-Part Study Pack artifact and File Quality Card
I. Mastery Check / Boss Battle	PASS	Part 9: The Context Master Boss Battle challenge

Report compiled and verified by PlayIQ Reconstruction Lane.